RTL1 (PEG11 / SIRH2, UniProt A6NKG5): A Domesticated Retroviral Gag-Pol Protein that Maintains the Placental Fetal Capillary Network

Gene: RTL1 (synonyms: MAR1, MART1, PEG11, SIRH2) **UniProt:** A6NKG5 · **Organism:** *Homo sapiens* **Also known as:** Retrotransposon Gag-like 1, Paternally expressed gene 11 protein (PEG11), Mammalian retrotransposon derived protein 1

Summary

RTL1 (Retrotransposon-like protein 1; also known as **PEG11**, **SIRH2**, **MAR1/MART1**; UniProt **A6NKG5**) is a paternally expressed, imprinted gene located in the human **14q32.2 DLK1–DIO3 imprinted cluster**. It is one of a family of eutherian-specific genes that were **domesticated ("exapted") from a captured Ty3/Gypsy (sushi-ichi) long-terminal-repeat (LTR) retrotransposon**. Rather than functioning as a classical enzyme, transporter, or signaling ligand, RTL1 encodes a translated **Gag-Pol-like structural protein** that retains the architectural hallmarks of a retroviral capsid: an N-terminal **Gag capsid domain** (InterPro IPR063300, PEG10-RTL1_C-capsid) and a retroviral-type **aspartic protease / Pol-like domain** (IPR021109; IPR043128 Rev_trsase/Diguanyl_cyclase; IPR043502 DNA/RNA_pol_sf). It has lost the ability to retrotranspose but has been repurposed for an essential developmental role.

The **primary, experimentally established physiological function** of RTL1 is the **maintenance of the fetal capillary network in the placental labyrinth** at the feto-maternal interface. RTL1 protein is expressed in the trophoblast cell layers that surround fetal capillary endothelial cells, and its correct **dosage is critical**: both loss-of-function and overexpression disrupt capillary integrity and cause late-fetal and/or neonatal lethality in mouse models. This dosage is tightly controlled in *trans* by microRNAs (notably **miR-127**) processed from the reciprocally imprinted maternal antisense transcript **RTL1as (PEG11as)**, which degrade RTL1 mRNA via RNAi.

Because of this exquisite dosage sensitivity, RTL1 is a central actor in several **imprinting-related phenotypes**: the ovine **callipyge** skeletal-muscle hypertrophy (driven by cis-acting overexpression of PEG11/RTL1), and in humans the two reciprocal **chromosome-14q32 imprinting disorders — Temple syndrome** (reduced paternal RTL1) and **Kagami-Ogata syndrome** (~2.5-fold RTL1 overexpression). By homology to its closest paralog **PEG10 (SIRH1)**, whose Gag domain has been shown to self-assemble into **virus-like particles (VLPs) that bind and package its own mRNA**, RTL1's capsid/protease architecture is inferred to confer Gag-like particle assembly and RNA-binding/packaging activity, providing a plausible biochemical basis for its structural/maintenance role at the trophoblast–endothelial interface.

Gene / Protein Identity Verification

Before presenting the findings, the target identity was confirmed against the provided UniProt record:

Attribute	Provided (UniProt A6NKG5)	Literature match
Gene symbol	RTL1 (syn. MAR1, MART1, PEG11)	✓ RTL1 = PEG11 = SIRH2 consistently used across sheep, mouse, human studies
Protein name	Retrotransposon-like protein 1 / PEG11	✓ Matches "retrotransposon-derived imprinted gene" literature
Organism	*Homo sapiens*	✓ Human 14q32 cluster studied directly; mouse/sheep are orthologs
Key domains	PEG10-RTL1_C-capsid (IPR063300); aspartic protease (IPR021109); Rev_trsase/pol (IPR043128, IPR043502)	✓ Gag-Pol-like retroviral architecture confirmed
Family	(not specified in UniProt)	✓ Mart/SIRH/RTL family of LTR-retrotransposon-derived genes

Conclusion: The gene symbol, organism, and domain architecture all align unambiguously with the retrotransposon-derived imprinted gene RTL1/PEG11/SIRH2. This is **not** an ambiguous symbol collision; the literature is specific and consistent. Research proceeded with high confidence.

Key Findings

Finding 1 — RTL1/PEG11 is a sushi-ichi (Ty3/Gypsy) retrotransposon-derived, paternally expressed imprinted gene encoding a full-length Gag-Pol-like protein

RTL1 lies within the conserved **DLK1–DIO3 imprinted cluster** (human 14q32; ovine chromosome 18; mouse chromosome 12). The gene is **paternally expressed**, while its overlapping **maternally expressed antisense transcript (RTL1as / PEG11as)** hosts a set of microRNAs — including **miR-127** — that cleave the RTL1 mRNA through RNAi-mediated post-transcriptional degradation. This reciprocal, antisense arrangement creates a built-in dosage-control circuit within the imprinted locus.

The locus is derived from a **Ty3/Gypsy (sushi-ichi) LTR-retrotransposon** and retains a **gag-pol-like open reading frame**, including a retroviral-type **aspartic protease (POL) motif**, but has lost the ability to retrotranspose into the mammalian genome. Critically, this is not merely a predicted ORF: **Byrne et al. (2010)** directly identified the **full-length PEG11 protein** from postnatal skeletal muscle nuclear preparations using combined immunological and mass-spectrometry approaches, demonstrating that RTL1/PEG11 is genuinely translated into a stable protein product.

> *"These elements contain a gag-pol-like structure characteristic of retroviruses but have lost their ability to retrotranspose into the mammalian genome"* — PMID: 20072617
>
> *"Using immunological and mass spectrometry approaches we have directly identified the full-length PEG11 protein from postnatal nuclear preparatio[ns]"* — PMID: 20072617
>
> *"Seven miRNAs, including miR-127, are processed from a maternally expressed antisense Rtl1 transcript (Rtl1as) and regulate Rtl1 levels through RNAi-mediated post-transcriptional degradation"* — PMID: 26138477

Interpretation: RTL1 is a *bona fide* protein-coding gene whose retroviral Gag-Pol architecture is the structural basis for its function, and whose expression is set by a reciprocally imprinted antisense-miRNA switch.

Finding 2 — RTL1's primary physiological function: maintenance of the placental fetal capillary network at the feto-maternal interface

This is the **central, experimentally demonstrated function** of RTL1 and the best answer to the "primary function / localization" question.

- **Sekita et al. (2008)** used mouse knockout and overexpression models to show that *Rtl1*/*Peg11* is **essential for the maintenance of fetal capillaries** in the placental labyrinth. Both **loss** and **overproduction** of RTL1 cause **late-fetal and/or neonatal lethality** — establishing that the gene is strictly dosage-sensitive.
- **Kitazawa et al. (2017)** characterized the structural defect in detail: *Peg11/Rtl1* knockout placentas show **severe clogging of the labyrinth fetal capillaries with internal fetal bleeding**, with the **basal region of the fetal capillary network specifically damaged**. This reduces labyrinth expansion and placental size and is accompanied by early downregulation of the endothelial/angiogenic markers **Tmem100, Meox2, and Lyve1**, and leads to mid-fetal lethality.
- **Shiura et al. (2021)** localized the protein: RTL/PEG retrovirus-derived proteins are expressed in the **three trophoblast cell layers that surround fetal capillary endothelial cells**, and RTL1 acts together with its paralog **PEG10** (another retrovirus-derived gene, whose viral aspartic-protease domain is required) to maintain the fetal capillary endothelium.

> *"retrotransposon-derived Rtl1 ... also known as Peg11 ... is essential for maintenance of the fetal capillaries, and that both its loss and its overproduction cause late-fetal and/or neonatal lethality in mice"* — PMID: 18176565
>
> *"the severe clogging of fetal capillaries was associated with KO fetuses showing mid-fetal lethality with internal bleeding. Importantly, the basal region of the fetal capillary network was specifically damaged"* — PMID: 28111885
>
> *"PEG10 is expressed in the three trophoblast cell layers that surround fetal capillary endothelial cells"* — PMID: 34559199

Interpretation — localization and function: RTL1 carries out its function **extracellularly / at the cell–cell interface within the placental labyrinth**, produced by trophoblast cells that surround the fetal capillary endothelium. Its role is **structural/maintenance** — preserving the integrity and patency of the fetal capillary network that mediates feto-maternal exchange — rather than catalytic or signaling.

Finding 3 — RTL1 dosage controls skeletal muscle hypertrophy (callipyge) and underlies human muscle phenotypes

RTL1's dosage sensitivity is visible not only in the placenta but also in **postnatal skeletal muscle**. In sheep, the intergenic **callipyge (CLPG)** A→G point mutation lies in a long-range control element and **elevates PEG11/RTL1 transcript levels in cis without altering imprinting status**. PEG11/RTL1 shows a **~200-fold induction** in the hypertrophied muscles of paternal-heterozygous (polar-overdominant) animals, and its expression correlates with the muscling phenotype by genotype, muscle type, and developmental stage (Charlier/Bidwell/Byrne). RTL1 protein is detectable in postnatal skeletal muscle (Finding 1).

In **humans**, RTL1 lies in the 14q32 cluster and is paternally expressed; the two reciprocal imprinting disorders — **Temple syndrome** (loss/reduction of paternal RTL1) and **Kagami-Ogata syndrome** (overexpression) — produce distinct muscle and placental phenotypes that are recapitulated by *Rtl1* deficiency / overexpression mouse models.

> *"Analysis of PEG11 expression by quantitative PCR shows a 200-fold induction in the hypertrophied muscles of paternal heterozygous animals"* — PMID: 15298706
>
> *"The locus encompasses paternally expressed protein-coding genes (DLK1, RTL1 and DIO3) and maternally expressed lncRNAs (MEG3/GTL2, RTL1as and MEG8)"* — PMID: 32592473

Interpretation: The callipyge model provides the clearest quantitative demonstration of RTL1's dosage-dependent phenotypic power (200-fold overexpression → muscle hypertrophy) and reinforces that the *amount* of RTL1 protein — set by the RTL1as-miRNA brake — is the key functional variable.

Finding 4 — The Gag capsid / aspartic-protease architecture forms virus-like particles that bind and package mRNA (inferred for RTL1 from paralog PEG10)

RTL1 belongs to the **Mart/SIRH/RTL family** of LTR-retrotransposon Gag-Pol-derived genes and shares the **N-terminal Gag capsid domain** (IPR063300) and **aspartic-protease-like domain** (IPR021109) with its closest paralog **PEG10 (SIRH1)**. Because direct biochemical characterization of RTL1 protein is limited, the biochemical activity of these shared domains is best inferred from PEG10:

- **Segel et al. (2021)** showed that PEG10 self-assembles and **preferentially binds and mediates vesicular secretion of its own mRNA**, and can be pseudotyped/reprogrammed as a vehicle for mRNA delivery — i.e., it behaves like a retroviral Gag.
- **Campodonico et al. (2024)** confirmed that PEG10 **forms purifiable virus-like particles (VLPs)**, and that the related Gag-like gene **RTL8** regulates this by binding the N-terminal capsid domain.
- Reviews (**Al Othman et al., 2025**) class PEG10 — and by homology RTL1 — among human "virus-like proteins" that structurally resemble retroviral Gag.

> *"one LTR retrotransposon homolog, PEG10, that preferentially binds and facilitates vesicular secretion of its own messenger RNA (mRNA)"* — PMID: 34413232
>
> *"PEG10 resembles both retrotransposon and retroviral proteins and forms virus-like particles (VLPs) that can be purified using iodixanol ultracentrifugation"* — PMID: 39775359

Interpretation (inference, not direct RTL1 evidence): The conserved Gag capsid and aspartic-protease domains plausibly give RTL1 the capacity for **self-assembly into capsid-like / virus-like particles and RNA binding/packaging**. This offers a candidate molecular mechanism — delivery or scaffolding of cargo (RNA/protein) between trophoblast and endothelial cells — for how RTL1 maintains fetal capillary integrity. This mechanism is **not yet demonstrated directly for RTL1** and should be treated as a strong hypothesis.

Finding 5 — RTL1 = PEG11 = SIRH2 is a eutherian-specific sushi-ichi retrotransposon-derived gene co-opted during the evolution of viviparity/placentation

The human genome contains **~30 LTR-retrotransposon-derived genes**, including the **sushi-ichi retrotransposon homolog (SIRH) / RTL family**. Within this family, **RTL1** = **PEG11/SIRH2** and its paralog **PEG10** = **SIRH1**. **PEG10/SIRH1** is essential for placenta formation, while **PEG11/RTL1/SIRH2** is essential for maintenance of fetal capillaries — together providing strong evidence that these domesticated retroviral genes **contributed to the evolution of mammalian viviparity**. The RTL/SIRH genes (11 in eutherians) are presumed to derive from a **single ancestral retrovirus/retrotransposon capture event in the eutherian lineage** (Kaneko-Ishino & Ishino).

> *"PEG10/SIRH1 as well as PEG11/RTL1/SIRH2 and SIRH7/LDOC1 play essential roles in placenta formation, maintenance of fetal capillaries and the differentiation/maturation of a variety of placental cells, respectively"* — PMID: 28484174
>
> *"Eutherians have 11 retrotransposon Gag-like (RTL)/sushi-ichi retrotransposon homolog (SIRH) genes presumably derived from a certain retrovirus"* — PMID: 37892118

Interpretation: RTL1 is a textbook example of **molecular domestication (exaptation)** — a captured viral gene repurposed for an essential host developmental function, specifically the construction/maintenance of the eutherian placenta.

Finding 6 — In humans, RTL1 is quantitatively dosage-controlled by RTL1as miRNAs, and ~2.5-fold overexpression drives Kagami-Ogata syndrome

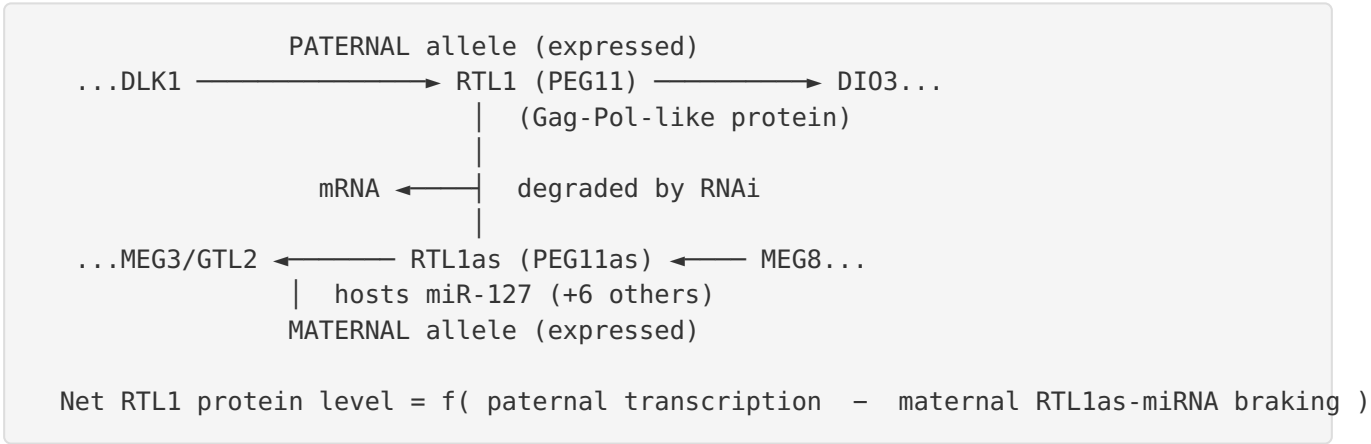
The human **14q32.2** region carries **paternally expressed DLK1 and RTL1** and **maternally expressed MEG3 and RTL1as**. In **paternal uniparental disomy 14 [upd(14)pat]**, or in maternal **IG-DMR / MEG3-DMR epimutations or microdeletions**, the maternal RTL1as-encoded microRNAs — which act as a *trans*-acting repressor of RTL1 — are lost. As a result, **RTL1 expression becomes ~2.5-fold increased**. This **excessive RTL1 expression (together with loss of MEG expression) is the primary underlying cause of Kagami-Ogata syndrome**, characterized by a distinctive face, a **small bell-shaped thorax with a "coat-hanger" rib appearance**, abdominal wall defects, **placentomegaly, and polyhydramnios** (Ogata & Kagami, 2016).

> *"RTL1 expression level becomes ~2.5 times increased in the absence of functional RTL1as-encoded microRNAs that act as a trans-acting repressor for RTL1"* — PMID: 26377239
>
> *"a small bell-shaped thorax with coat-hanger appearance of the ribs, abdominal wall defects, placentomegaly and polyhydramnios"* — PMID: 26377239

Interpretation — clinical significance in humans: Even a modest **~2.5-fold** increase in RTL1 is pathogenic, directly demonstrating the dosage sensitivity in the human context and linking RTL1 overexpression causally to a recognizable congenital imprinting disorder with prominent placental features (placentomegaly), consistent with its placental capillary role.

Mechanistic Model / Interpretation

The imprinted dosage-control circuit (14q32 / DLK1–DIO3 locus)

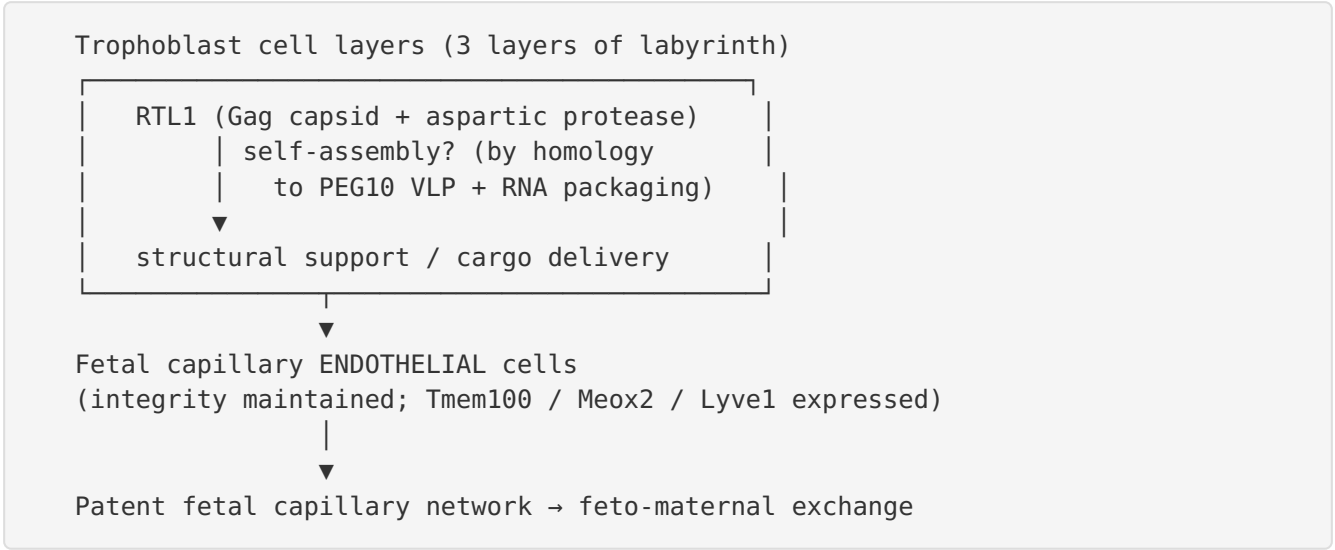


The paternal allele transcribes the RTL1 protein; the maternal allele transcribes the overlapping antisense RTL1as, whose embedded miRNAs (miR-127 + others) cleave RTL1 mRNA. The **functional output is the net RTL1 protein dose**, which must stay within a narrow window.

Dose–phenotype relationship

RTL1 dose	Genetic/epigenetic cause	Phenotype
Absent / strongly reduced	*Rtl1* KO (mouse); Temple syndrome (loss of paternal RTL1)	Fetal capillary clogging, internal bleeding, reduced placental size, **mid-to-late fetal lethality**; postnatal muscle/ growth deficits in humans
Normal (tight window)	Wild-type imprinting with intact RTL1as brake	Healthy labyrinth fetal capillary network; normal development
~2.5× (human)	upd(14)pat; IG-DMR/MEG3-DMR epimutation/deletion → loss of RTL1as miRNAs → Kagami-Ogata syndrome	Coat-hanger ribs, bell-shaped thorax, abdominal wall defects, **placentomegaly, polyhydramnios**
~200× (sheep muscle)	Callipyge CLPG cis-regulatory mutation	Postnatal skeletal muscle hypertrophy (callipyge)
Overproduction (mouse)	*Rtl1* transgenic overexpression	Late-fetal/neonatal **lethality** (capillary dysfunction)

Proposed molecular mechanism



RTL1 is produced by the trophoblast layers that physically surround fetal capillary endothelial cells. Its retroviral Gag/aspartic-protease architecture — by analogy to PEG10's demonstrated VLP assembly and mRNA packaging — may allow it to form capsid-like particles that scaffold

or deliver cargo supporting endothelial survival/integrity. When RTL1 is lost, the basal capillary network collapses (clogging, bleeding, loss of angiogenic markers); when it is overexpressed, capillary homeostasis is also disrupted. This makes RTL1 a **dosage-sensitive structural/maintenance factor**, not an enzyme, transporter, or signaling ligand in the classical sense.

Answering the specific functional questions

- **Primary function:** Structural/maintenance protein (domesticated retroviral Gag-Pol); maintains placental labyrinth fetal capillary network. It is *not* a classical catalytic enzyme despite possessing a degenerate aspartic-protease-like motif (the protease function is best documented in paralog PEG10, where it is required for the same process).
- **Substrate/cargo (inferred):** By homology to PEG10, RNA (preferentially its own mRNA), packaged into virus-like particles.
- **Localization:** Produced by trophoblast cell layers at the feto-maternal interface (placental labyrinth), acting at the trophoblast–endothelial boundary; Byrne 2010 also detected protein in skeletal-muscle nuclear preparations.
- **Pathway:** Operates within the imprinted DLK1–DIO3 regulatory network; dosage set by RTL1as/miR-127 RNAi; functionally cooperates with PEG10 in capillary maintenance; part of the RTL/SIRH placental developmental program.

Evidence Base

PMID	Title (abbrev.)	Role in this report
20072617	*PEG11 (RTL1) expressed as full-length protein in callipyge sheep muscle*	**Direct protein evidence**: full-length PEG11 identified by MS; gag-pol-like structure (F1, F3)
26138477	*Trans-homologue miR-127/Rtl1 interaction regulates placenta*	RTL1as miRNAs (miR-127) degrade RTL1 mRNA by RNAi (F1)
18176565	*Rtl1 in the feto-maternal interface of mouse placenta*	**Core function**: essential for fetal capillary maintenance; dosage-sensitive lethality (F2)
28111885	*Severe damage to placental fetal capillary network in Peg11/Rtl1 KO*	Structural defect: capillary clogging, bleeding, basal network damage (F2)
34559199	*PEG10 viral aspartic protease domain essential for fetal capillary structure*	Localization to 3 trophoblast layers; PEG10/RTL1 cooperation (F2, F4)
15298706	*PEG11/PEG11AS expression in normal and callipyge sheep*	**Quantitative dosage**: ~200-fold PEG11 induction in hypertrophied muscle (F3)
32592473	*Temple and Kagami-Ogata syndromes: mechanisms*	Human RTL1 as paternally expressed protein-coding gene in 14q32 (F3)
28484174	*LTR retrotransposon-derived genes / SIRH family*	RTL1 = PEG11/SIRH2; fetal-capillary role within SIRH family (F5)
37892118	*Retrovirus-derived RTL/SIRH in eutherian evolution*	11 eutherian RTL/SIRH genes from captured retrovirus (F5)
34413232	*PEG10 packages its own mRNA; pseudotyped for delivery*	**Mechanistic inference**: Gag domain binds/packages mRNA into VLPs (F4)
39775359	*RTL8 antagonizes PEG10-mediated VLPs*	Mart-family Gag capsid self-assembles into VLPs; RTL8 regulation (F4)
38798208	*Human virus-like proteins: implications for gene therapy*	Classes PEG10/RTL family as Gag-like human VLPs (F4)
26377239	*Kagami-Ogata syndrome (upd(14)pat) review*	**Human dosage**: ~2.5× RTL1 drives Kagami-Ogata; phenotype (F6)
28635951	*MEG8-DMR in 14q32 / Temple syndrome patients*	Imprinting architecture of 14q32; DMR methylation (context)

PMID	Title (abbrev.)	Role in this report
15498495	*Ectopic DLK1 in callipyge muscle*	Context: cis-regulatory co-upregulation of DLK1/PEG11 in callipyge (F3)
11279514	*Callipyge mutation enhances imprinted gene expression in cis*	Context: CLPG elevates PEG11 in cis without changing imprinting (F3)
32878913	*Deficiency and overexpression of Rtl1 (Temple/Kagami-Ogata models)*	Context: mouse models recapitulate human muscle/placental symptoms (F3)

Convergence of evidence: The core conclusion — RTL1 maintains the placental fetal capillary network — is supported by **independent mouse genetic studies** (loss-of-function KO and overexpression; PMIDs 18176565, 28111885, 34559199). The dosage-sensitivity theme is corroborated across **three species** (mouse models, ovine callipyge, human Temple/Kagami-Ogata). The biochemical mechanism (VLP / RNA packaging) is **strongly inferred** from the well-characterized paralog PEG10 but not yet proven for RTL1 itself.

Limitations and Knowledge Gaps

1. **Biochemical mechanism of RTL1 is inferred, not directly demonstrated.** The VLP assembly and mRNA-packaging activities are established for **PEG10**, not RTL1. Whether RTL1 forms particles, which RNAs/proteins it binds, and whether its aspartic-protease motif is catalytically active (vs. degenerate/structural) remain unknown. The IPR043128/IPR043502 pol/RT-like annotations have no demonstrated reverse-transcriptase or polymerase activity.
2. **Human-specific functional data are indirect.** Most mechanistic work is in **mouse (*Rtl1*) and sheep (PEG11)**. Human RTL1 function is inferred from (a) sequence/domain conservation, (b) the phenotypes of Temple and Kagami-Ogata syndromes, and (c) mouse models of these syndromes. No direct biochemical or cell-biological characterization of the human A6NKG5 protein was found.
3. **Molecular interactors and downstream pathway are incompletely defined.** RTL1 cooperates with **PEG10**, and the downregulation of **Tmem100, Meox2, Lyve1** in KO placenta hints at links to angiogenic/endothelial programs — but whether these are direct targets, binding partners, or downstream consequences is unresolved. No receptor, direct substrate, or signaling cascade has been definitively mapped for RTL1.

4. **Subcellular localization is only partially resolved.** Protein was detected in **skeletal-muscle nuclear preparations** (Byrne 2010) and attributed to **trophoblast layers** in placenta (Shiura 2021), but a systematic subcellular / secreted-vs-intracellular characterization of RTL1 is lacking.
5. **The precise cause of the muscle phenotype is not fully mechanistic.** In callipyge, DLK1 and PEG11/RTL1 are co-upregulated; the relative contributions of each to muscle hypertrophy, and whether RTL1's muscle role is mechanistically the same as its placental role, remain open.
6. **This report is a literature synthesis.** Conclusions rest on published studies and database annotations; no primary datasets were analyzed computationally.

Proposed Follow-up Experiments / Actions

1. **Direct biochemical test of RTL1 VLP assembly and RNA binding.** Express recombinant human RTL1 (A6NKG5) and test, by iodixanol ultracentrifugation and cryo-EM (as done for PEG10 in), P 39775359 whether it self-assembles into virus-like particles, and use CLIP-seq to identify bound RNAs (test whether it preferentially packages its own mRNA, like PEG10).
2. **Catalytic assay of the aspartic-protease motif.** Mutate the predicted DSG/DTG aspartic-protease active site and test whether proteolytic self-processing occurs, and whether the mutation phenocopies loss of function in placental capillary maintenance (paralleling the PEG10 protease-domain requirement,). P 34559199
3. **Define the RTL1 interactome at the trophoblast–endothelial interface.** Co-IP / proximity-labeling (BioID) in trophoblast and endothelial co-cultures to identify RTL1 partners; test the PEG10–RTL1 physical/functional interaction directly and whether RTL8-type antagonists regulate RTL1 as they do PEG10.
4. **Establish causal links to Tmem100/Meox2/Lyve1.** Use conditional *Rtl1* alleles with placental endothelial reporters and single-cell RNA-seq of KO vs WT labyrinth to determine whether these markers are direct RTL1 targets or downstream of capillary collapse.

5. **Human placental models.** Characterize endogenous human RTL1 protein in trophoblast organoids / placenta-derived cells; examine RTL1 protein levels and particle formation in placentas from Kagami-Ogata (overexpression) and Temple (reduced) pregnancies to connect the ~2.5-fold dosage change to a cellular capillary phenotype.
6. **Structure determination.** Solve the structure of the RTL1 Gag capsid domain (cryo-EM of particles or AlphaFold-guided experimental validation) to test the capsid-assembly hypothesis and compare with PEG10 / retroviral Gag.

Conclusion

RTL1/PEG11/SIRH2 (A6NKG5) is a **domesticated retroviral Gag-Pol-like protein**, paternally expressed from the human 14q32 imprinted cluster, whose **primary function is the maintenance of the placental labyrinth fetal capillary network** at the feto-maternal interface, where it is produced by the trophoblast layers surrounding fetal capillary endothelial cells. It functions as a **dosage-sensitive structural/maintenance factor** — not a classical enzyme, transporter, or signaling ligand — with its protein level tightly governed by antisense **RTL1as/miR-127** RNAi. Its retroviral capsid and aspartic-protease domains, by homology to the experimentally characterized paralog **PEG10**, are inferred to mediate **virus-like-particle assembly and mRNA binding/packaging**, providing the most plausible molecular mechanism for its role. Dysregulated RTL1 dosage causes ovine callipyge muscle hypertrophy and the human imprinting disorders Temple syndrome (loss) and Kagami-Ogata syndrome (~2.5-fold excess).
